

JKBOSE



2026 CLASS 11th



Baaz Batch
Imdabad.

SOME BASIC CONCEPT OF CHEMISTRY

CHEMISTRY

LECTURE-3

BY-VISHAL SIR

TOPICS TO BE COVERED

1 Laws of Chemical Combination

(Numericals)

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Me Sawaal Aata
Hai

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REVISION

Matter.

Physical Classification

- a) Solid
- b) Liquid
- c) Gas.

Chemical Classification

Mixture

Homogeneous

Heterogeneous

Pure Substance

Element

- a) Metal
- b) Non-metal
- c) Metalloids

Compound

Organic Inorganic

Homogeneous mixture, Heterogeneous mixture, Pure substance (Element or Compound)

① Dal → Heterogeneous mixture.

2. Tea → Heterogeneous mixture.

3. Cloud → Heterogeneous mixture.

4. IRON (Fe) → Element.

5. Water (H₂O) → Compound.

6. Blood → Heterogeneous mixture.

7. Air → Homogeneous mixture.

8. Alloy → Homogeneous mixture.

9. Milk → Heterogeneous mixture.

10. Salt solution → Homogeneous mixture.
NaCl + H₂O

11. Table salt → NaCl.
Pure substance

12. Tap water → (H₂O + minerals)
Homogeneous mixture

13. CaCO₃ → Compound
Homogeneous mixture

14. H₂SO₄ → Compound

15. Glucose.
C₆H₁₂O₆ → Compound.

Homogeneous mixture of Elements
↓
Alloy

Laws of Chemical Combination $\Rightarrow$

When elements combine in a chemical reaction, they follow some rules called laws of chemical combination.

Namely.

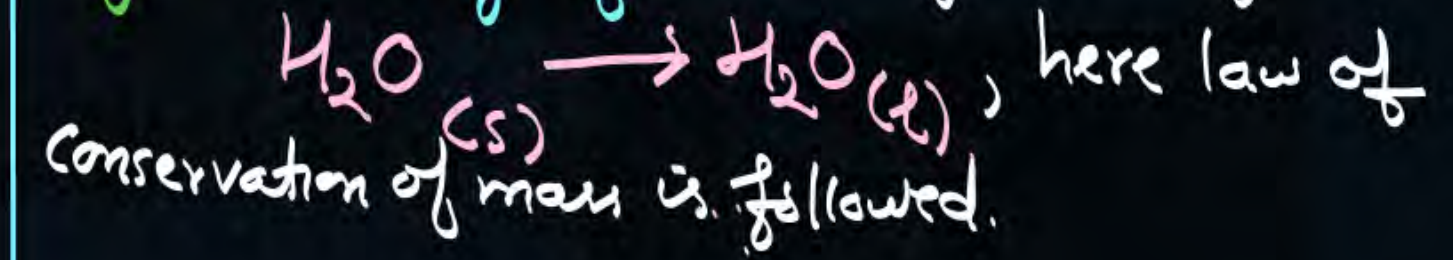
- 1) Law of conservation of mass.
- 2) Law of definite proportion or constant composition
- 3) Law of multiple proportion
- 4) Gay Lussacs law of Gases.
- 5) Avagadro law.

1) Law of conservation of mass $\Rightarrow$
It was given by Antoine Lavoisier.
(Father of Chemistry).

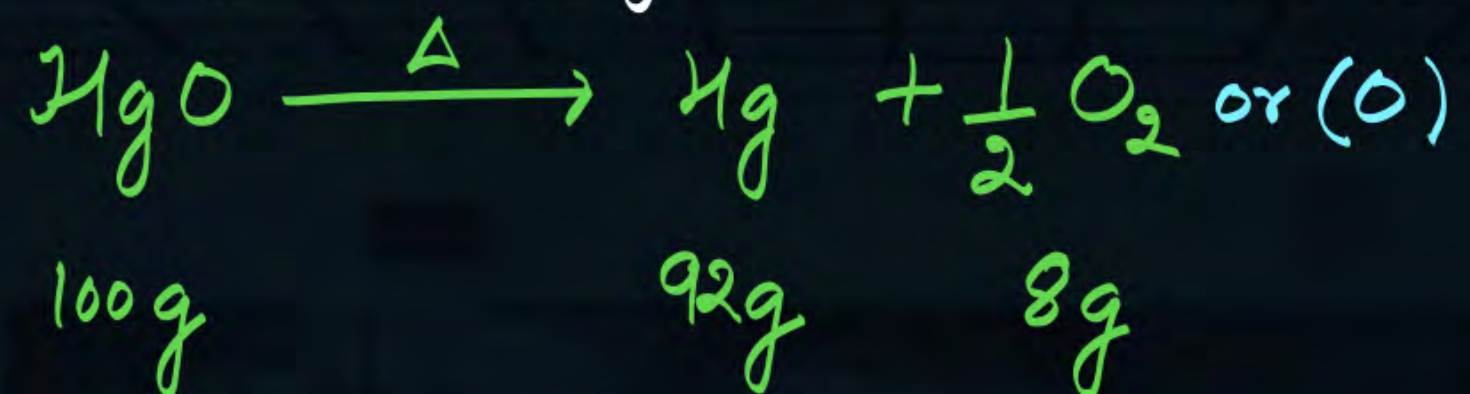
It states that "mass can be neither be created nor be destroyed"
OR.

In any physical or chemical change
sum of the mass of reactants is always equal to the sum of mass of products".

e.g i) Melting of Ice. (Physical change).



ii) Decomposition of HgO (Mercuric oxide).



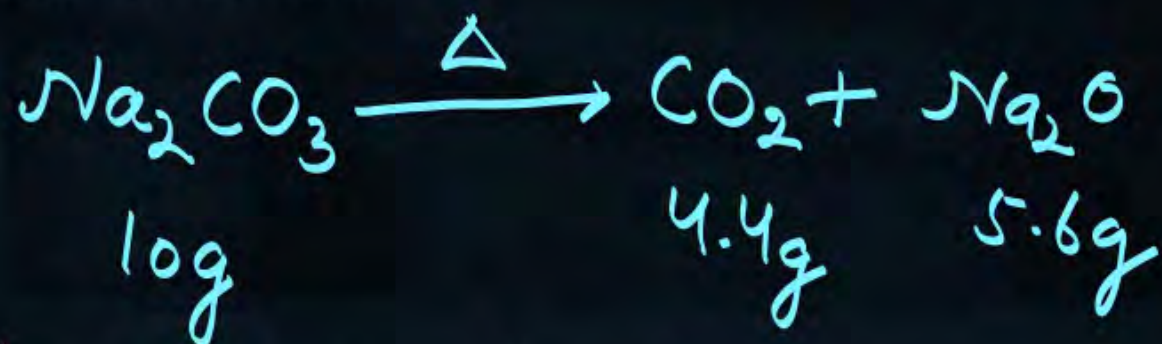
NOTE $\Rightarrow$ Law of conservation of mass is not applicable on nuclear reactions.

QUESTION



On heating, 10.0 grams of sodium carbonate (Na_2CO_3), 4.4 g of carbon dioxide (CO_2) and 5.6 g of sodium oxide (Na_2O) is produced. Show that this reaction obeys the law of conservation of mass.

Sol. $\Rightarrow$



According to law of conservation of mass.

Sum of mass of Reactants = Sum of mass of products

$$\log = 4.4\text{g} + 5.6\text{g}$$

$\log = \log$

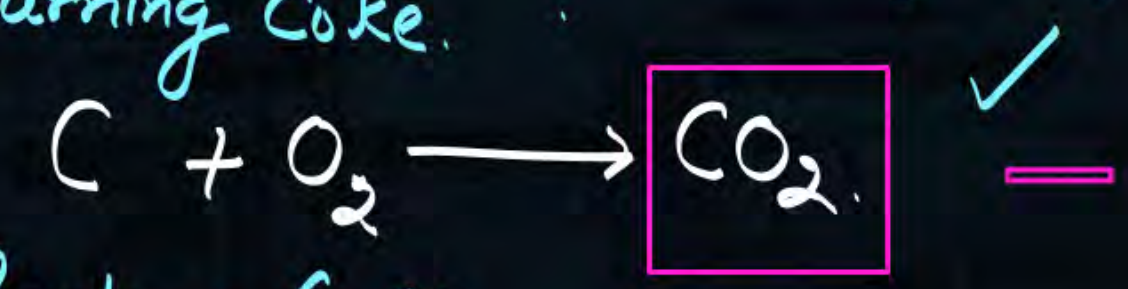
2). Law of definite proportion or constant composition.

Given by Joseph Proust.

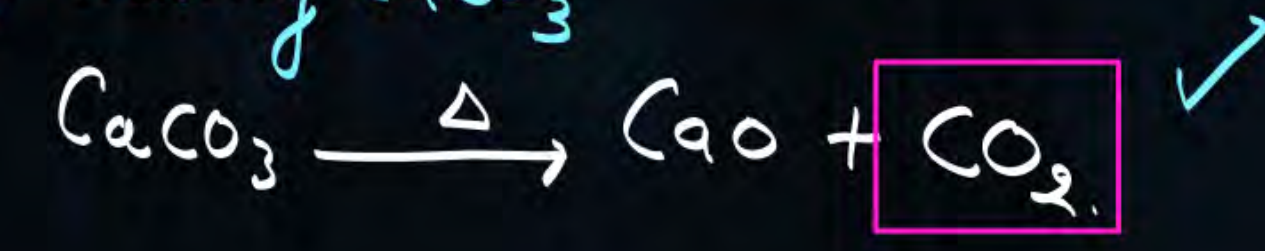
It states that "in any chemical substance (compound) elements are present in a fixed proportion by mass."

e.g. Carbon dioxide can be prepared by no. of ways.

i) By burning Coke.



ii) By heating $CaCO_3$



In both the case the ratio of mass of C + oxygen is fixed

$$12 : 32 \text{ or } \frac{12}{32} \times 3 \quad \text{i.e. } 3 : 8$$

Example no. 2. If we take sample of water from Chenab & Jehlum, the chemical formula is H_2O .

The ratio of mass of H:O comes out to be same for both sample. $2 : 16$ or $1 : 8$.

$C = 12g$
$O = 16g$
$H = 1g$

QUESTION

2.0 g of a metal burnt in oxygen gave 3.2 g of its oxide. 1.42 g of the same metal heated in steam gave 2.27 g of its oxide. Which law is shown by this data?

Sol: → Case 1.

Mass of metal oxide = 3.2 g
Mass of metal = 2.0 g
Mass of oxygen = 3.2 g - 2.0 g
= 1.2 g

Therefore Ratio of Mass of Metal & Oxygen

$$\frac{2.0}{1.2} = 1.6$$

Case 2.

Mass of metal oxide = 2.27 g
Mass of metal = 1.42 g
Mass of oxygen = 2.27 - 1.42
= 0.85 g

$$\text{Ratio of mass of metal \& oxygen} = \frac{1.42}{0.85} = 1.6$$

Yes, This data follow law of
Constant composition.

$$\begin{aligned} \text{Case 1} &= M/O = 1.6 \rightarrow x \\ \text{Case 2} &= M/O = 1.6 \rightarrow x \end{aligned}$$

QUESTION

Q4Q

Home work



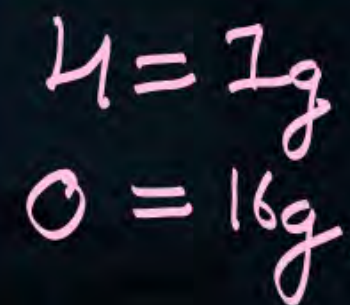
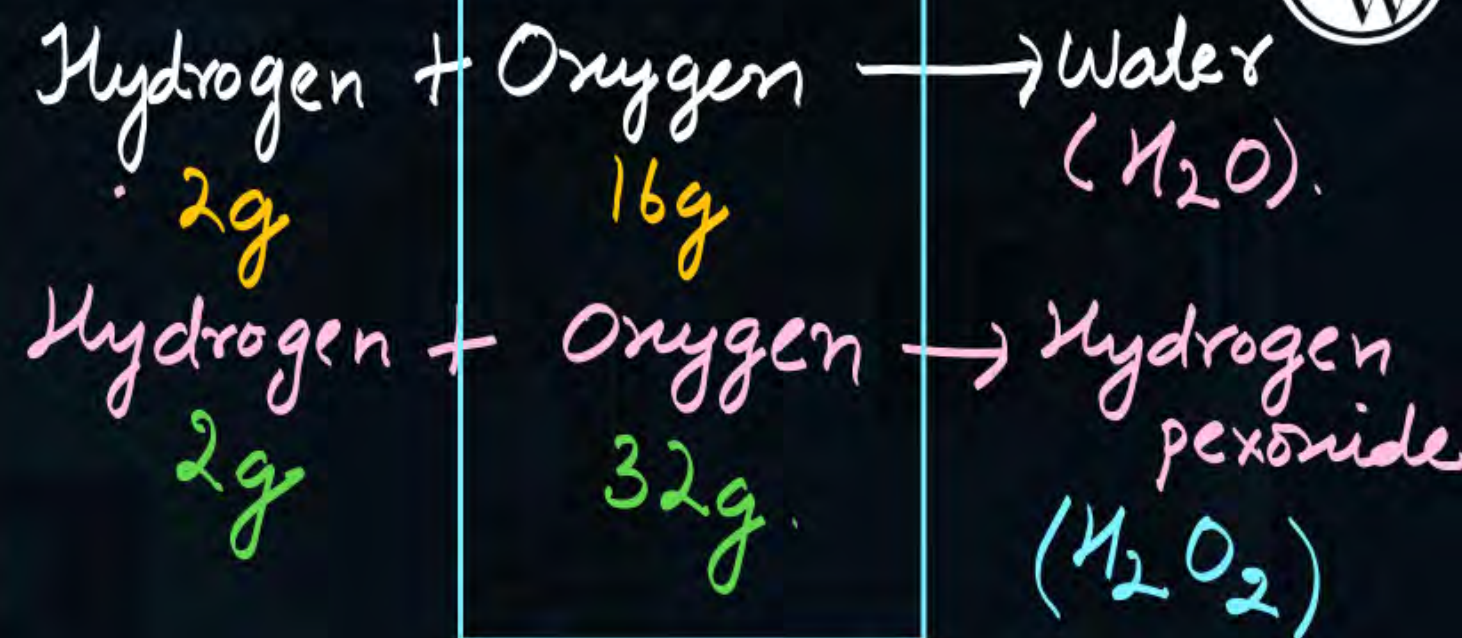
1. 3.75 g of cupric oxide was reduced by heating in a current of hydrogen and the weight of copper remained was 1.098 g. In another experiment, 1.179 g of copper was dissolved in HNO_3 and resulting copper nitrate converted into CuO by ignition the weight of CuO was 1.476 g. Show that these results, illustrates the law of constant composition.

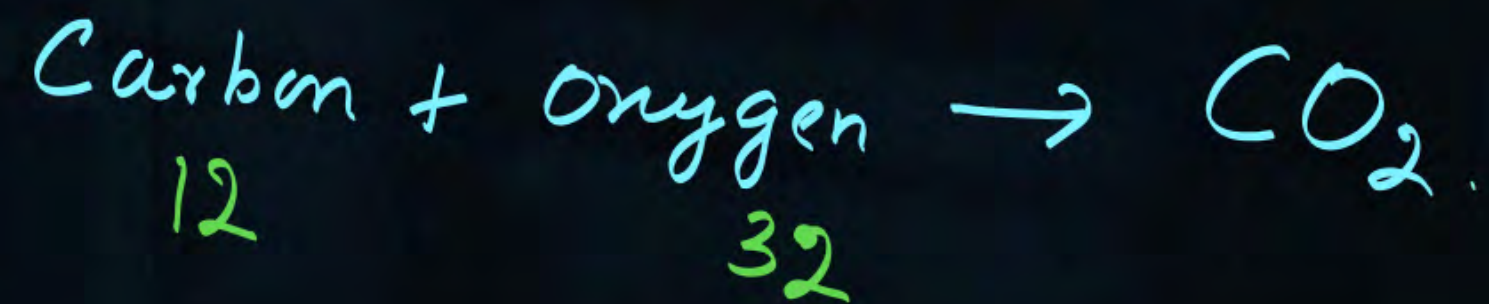
3) Law of multiple proportion →

Given by Dalton.

It states that "when two different elements combine to produce two or more different compounds, then mass of one element which combines with the fixed mass of other element in these compounds, bears a simple whole no. ratio by mass".

For example, Hydrogen & Oxygen combine to produce two different compounds.

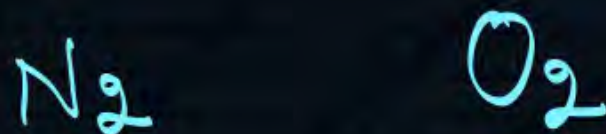




$$16 : 32$$

$$1 : 2$$

QUESTION



The following data are obtained when dinitrogen and dioxygen react together to form different compounds:

Mass of dinitrogen	Mass of dioxygen
14 g	16 g
14 g	32 g
28 g	32 g
28 g	96 g

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Which law of chemical combination is obeyed by the above experimental data?

1

Law of conservation of mass

2

Law of multiple proportions

3

Law of definite proportion

4

Avogadro's Law

QUESTION

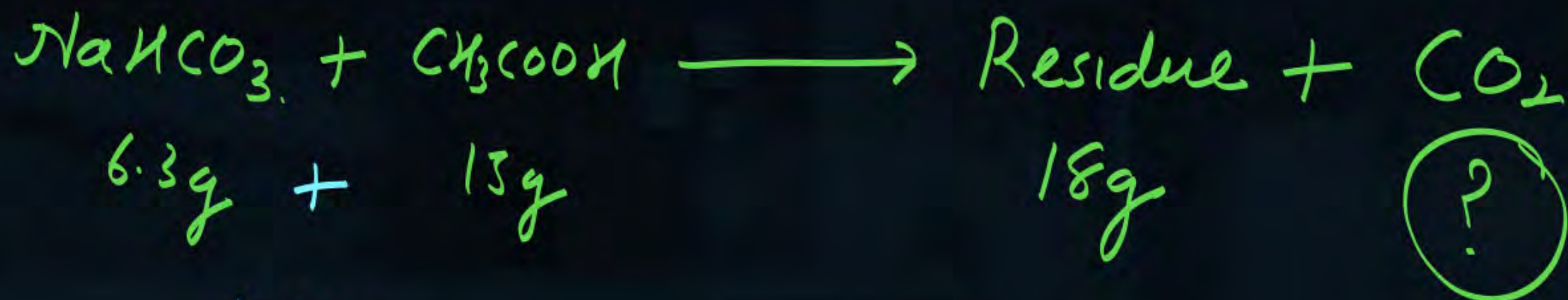
If 6.3 g of NaHCO_3 are added to 15.0 g CH_3COOH solution, the residue is found to weigh 18.0 g. What is the mass of CO_2 released in the reaction?

a) 2.3 g

b) 4.2 g

c) 10.3 g

d) 3.3 g



$$21.3\text{g} = 18\text{g} + x$$

$$21.3 - 18 = x$$

$$3.3$$

Law of
Conservation of mass

REVISE KAR K Aana.

Thank
You